

Exp. No.1b

CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

AIM:

To generate and analyze an Amplitude Modulated (AM) signal in MATLAB by creating the message and carrier signals, determining the modulation index, identifying the upper and lower sideband frequencies using FFT analysis, calculating the carrier, sideband, and total transmitted power, and studying the effect of modulation index on the AM signal characteristics and transmission efficiency.

APPARATUS REQUIRED:

- MATLAB SOFTWARE

Objective :

To generate an Amplitude Modulated (AM) signal in MATLAB using specified carrier and modulating signal parameters

PROGRAM 1:

```
clc; clear; close all;

Am = 1;

fm = 500;

Ac = 5;

fc = 5000;

fs = 20 * fc;

t = 0:1/fs:4/fm;

m = Am * sin(2 * pi * fm * t);

c = Ac * sin(2 * pi * fc * t);

mu = Am / Ac;

am = Ac * (1 + mu * sin(2 * pi * fm * t)) .* sin(2 * pi * fc * t);

dem = abs(hilbert(am));

dem = dem - mean(dem);

N = length(am);

f = linspace(-fs/2, fs/2, N);

AM_FFT = abs(fftshift(fft(am))) / N;

USB = fc + fm;

LSB = fc - fm;
```

$P_c = A_c^2 / 2;$

$P_{sb} = (\mu^2 * P_c) / 4;$

figure('Name', 'AM Modulation', 'NumberTitle', 'off');

subplot(3,2,1), plot(t, m), title('Message'), grid on

subplot(3,2,2), plot(t, c), title('Carrier'), grid on

subplot(3,2,3), plot(t, am), title('AM Signal'), grid on

subplot(3,2,4), plot(t, dem), title('Demodulated'), grid on

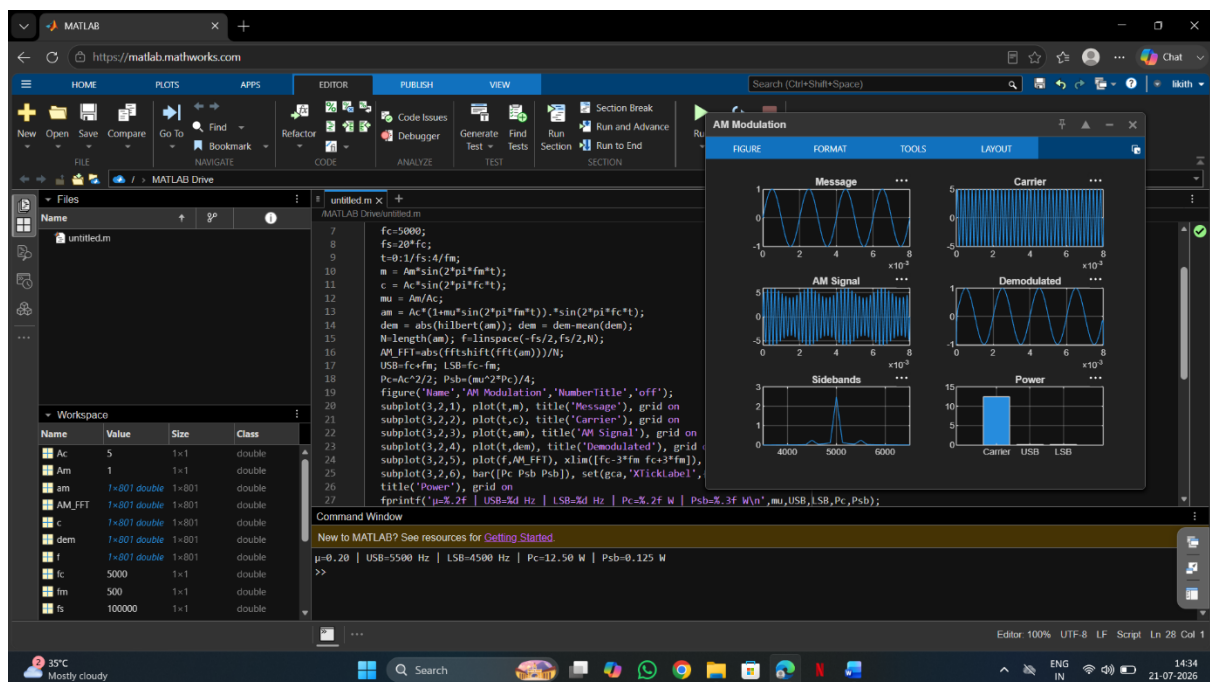
subplot(3,2,5), plot(f, AM_FFT), xlim([fc - 3 * fm fc + 3 * fm]), title('Sidebands'), grid on

subplot(3,2,6), bar([Pc Psb Psb]), set(gca, 'XTickLabel', {'Carrier', 'USB', 'LSB'}), title('Power'), grid on

fprintf('μ=%2f | USB=%d Hz | LSB=%d Hz | Pc=%2f W | Psb=%.3f W\n', ...

mu, USB, LSB, Pc, Psb);

OUTPUT (GRAPH)



RESULT :

The AM signal was successfully generated and analyzed in MATLAB.